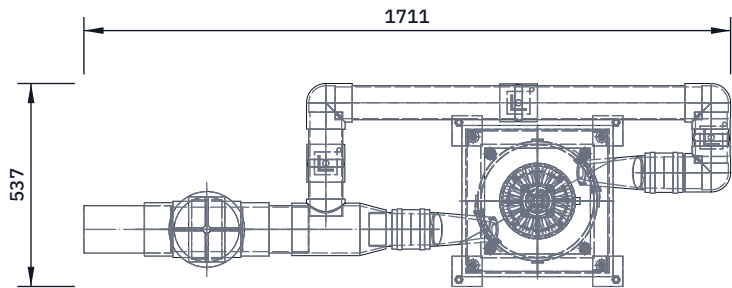
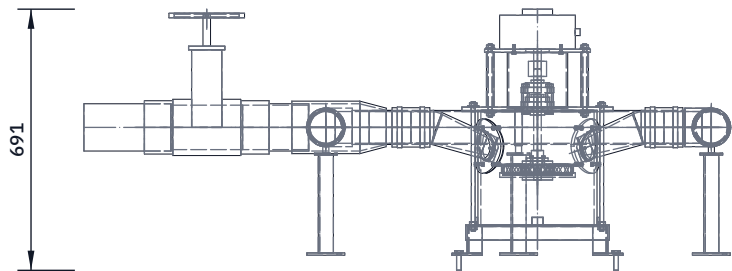


CONCEPT, NOT FOR FABRICATION

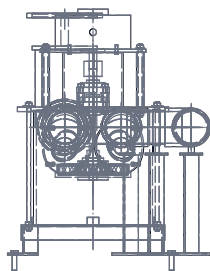
PicoFlow · github.com/BoujeeEnjinia1701/picoflow



TOP VIEW
Scale 1:20; looking down (along -Z)

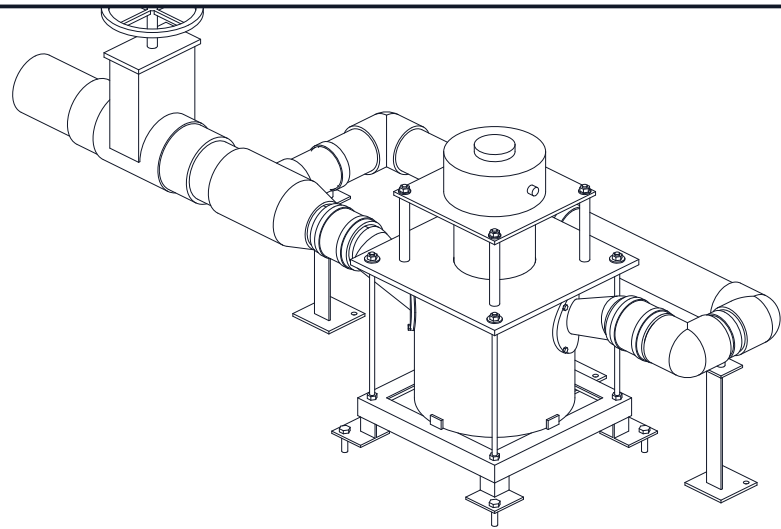


FRONT VIEW
Scale 1:20; looking at the front (along +Y)



RIGHT VIEW
Scale 1:20; looking at the right side (along -X)

REV	DESCRIPTION	DATE	BY
P1	General arrangement for TRL 3 (PCF-CAL-001)	2026-09-25	AC
P2	Penstock 125 mm, 33 mm inserts (PCF-DDR-002)	2026-09-25	AC
P3	Design for construction (PCF-DDR-003)	2026-10-01	AC
P4	Full-bore 125 mm inlet valve (PCF-DDR-003, A2)	2026-10-01	AC
P5	Frame steel welded bare, then galvanized or painted (2026-10-02)	2026-10-02	AC



ISOMETRIC VIEW

Not to scale; penstock shown as a stub; seen from the front right and above, about 30 deg elevation

MAIN DIMENSIONS (MM), Z FROM NORMAL TAILWATER

Nozzle centerline 280 above tailwater (R13: 300 max)
Runner 200 OD, 150 pitch dia., 20 buckets, mid-plane 224
Runner underside 199; housing 315 OD x 303, bottom at 80
Two jets, 34 mm inserts (20 to 45 range), 20 deg below runner plane
Jets strike the pitch circle at +Y and -Y, opposed
Shaft 20 dia. x 294; two UCF204 units on the lid at 392
Generator 200 dia. x 90 on plate at 537
Penstock 125 OD at Y -75; branches 90 OD at Z 338
Lid 400 sq. x 12; frame 380 sq., 4 tie rods

PARTS LIST (ITEMS MATCH BOM/BOM.CSV)

- 1 Turgo runner, printed
- 2 Shaft and hub
- 3 Bearing units, posts, plate, guard
- 4 Jaw coupling
- 5 Generator, low-speed BLDC
- 6 Housing and lid

PARTS LIST, CONTINUED

- 7 Manifold, two nozzles
- 8 Inlet gate valve, 125 mm full bore
- 9 Penstock (stub shown)
- 11 Turbine frame
- 18 Pipe stands; 19 Fixings
- Items 10, 12 to 17 not shown
- Guard the coupling; DC side 100 V

PICOFLOW

github.com/BoujeeEnjinia1701/picoflow

Turbine unit general arrangement

DRAWING NO. PCF-DWG-001		REV P5	SHEET 1 of 1	SIZE ANSI B
SCALE 1:20	UNITS mm	PROJECTION 	DATE 2026-10-02	DRAWN Amish Chadha
MATERIAL / NOTES Runner and nozzles PETG; housing PVC SN4; lid HDPE; frame steel, galvanized or painted after welding. PRELIMINARY, NOT FOR FABRICATION				
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